



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
14-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

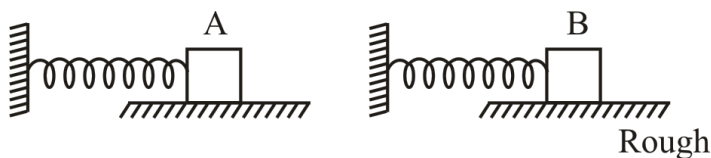
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

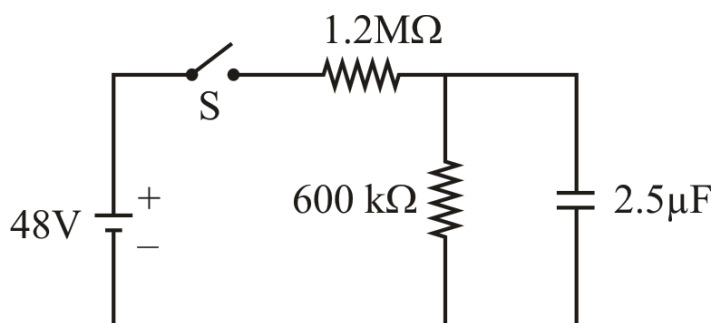
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider two identical oscillators of the type shown in figure. For one of the oscillators (say A), the surface is frictionless, but there is viscous damping $F_{\text{drag}} = -bv$. For the other (say B), the surface has coefficient of kinetic friction μ_k , but there is no viscous damping. The masses are both pulled away from equilibrium the same distance to the right and then released. When they first reach the natural length position, the magnitude of the viscous damping force for the first oscillator is equal to the magnitude of the frictional force for the second oscillator.



- (A) The time period of oscillation of A is more than the time period of oscillation of B.
- (B) A reaches farther from the natural length position than B at the first stop.
- (C) The kinetic energy of A at the natural length position is more than that for B.
- (D) The amplitude of A decreases exponentially with time.

2. A linearly polarized electromagnetic wave in vacuum $E = 3 \cos(2\pi(1.8z - 5.4 \times 10^8 t)) \hat{j}$ N / C is incident normally on a perfectly reflecting wall at $z = 0$ m. Choose the correct option.
- (A) The wavelength is $5/9$ m
- (B) The frequency of electromagnetic wave is 5.4×10^8 Hz.
- (C) The magnetic field of the reflected wave will be $-10^{-8} \cos(2\pi(1.8z + 5.4 \times 10^8 t)) \hat{i}$ T.
- (D) The electric field of the reflected wave will be $-3 \cos(2\pi(1.8z + 5.4 \times 10^8 t)) \hat{j}$ N / C.
3. In the circuit shown in figure capacitor was initially uncharged switch is closed at $t = 0$. Select correct alternatives.



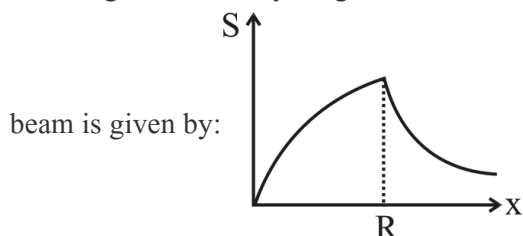
- (A) The initial battery current immediately after switch S is closed is $40 \mu\text{A}$.
- (B) The battery current, long time after switch S is closed is $\frac{80}{3} \mu\text{A}$.
- (C) The time after which current through capacitor branch becomes half is $\ln 4$ sec.
- (D) The current through the $600\text{k}\Omega$ as a function of time is $\frac{80}{3}(1 - e^{-t})\mu\text{A}$.

Space for Rough Work

4. Consider a long cylindrical uniform beam of protons moving with a velocity v each. The beam is of radius R . The total current due to the beam is I . Which of the following is correct?

(A) The Poynting vector is in the direction of v .

(B) The magnitude of Poynting vector as a function of distance from the axis of the cylindrical



(C) The rate of energy flow through the cross-section of the beam is $\frac{I^2 R^2}{8\pi\epsilon_0 v}$.

(D) The electric field at a distance x from the axis ($x < R$), is $\frac{Ix}{4\pi\epsilon_0 R^2 v}$.

5. The formula for the self-inductance of a ring was first given by Kirchoff. If the radius of the ring is a and the cross-section radius of its wire is b , the formula for its self-inductance is given by

$L = \mu_0 a \left(\ln \left(\frac{8a}{b} \right) - 2 \right)$. Take $\ln 160 = 5$. Suppose that we have two circular conductors of radius 4 cm and 2 cm. Their radii of cross section are 1 mm each. What can be their mutual inductance?

(A) $0.50 \mu\text{H}$

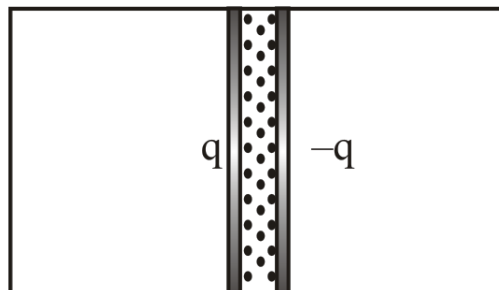
(B) $0.02 \mu\text{H}$

(C) $0.12 \mu\text{H}$

(D) $0.72 \mu\text{H}$

Space for Rough Work

6. We have two charged discs of radius r and charge q and $-q$ as shown in the figure. The region behind the plates is vacuum and the region in between the discs is filled with mixture of helium and hydrogen in mole ratio 1:1. The cylinder in which the system is located is perfectly adiabatic. If the charge on each plate is halved, comment on the parameters of the gas when steady state is reached. Assume that the process on gas is quasi static and reversible. The permittivity of gas is same as that of free space.



- (A) The pressure is one fourth the original pressure.
- (B) The temperature becomes $2^{\frac{-3}{2}}$ times.
- (C) The distance between the plates becomes $2^{\frac{4}{3}}$ times.
- (D) The energy stored in the capacitor becomes $2^{\frac{-2}{3}}$ times.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

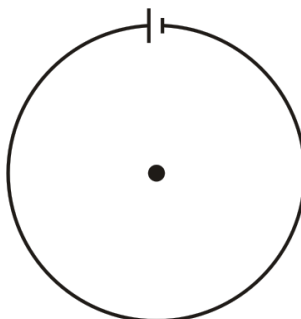
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A wire of resistivity $1 \text{ n}\Omega\text{-m}$, length 2m and area of cross-section 1mm^2 is rolled in the shape of a ring with ten turns. It is connected to a battery of voltage V as shown in the figure.

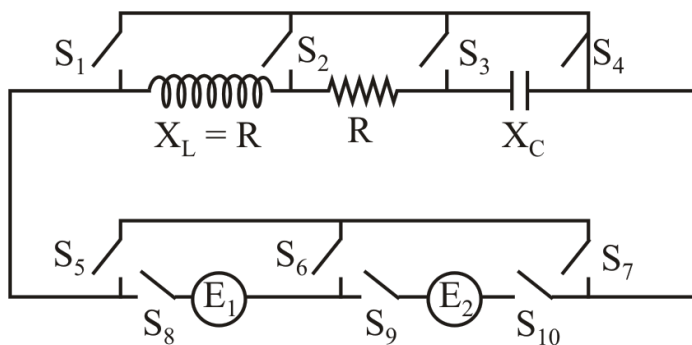


7. What is the value of V so that the magnetic field at the centre is 1T ?
 (A) 5 V (B) 10 V
 (C) 20 V (D) 2.5 V
8. What torque is needed to hold this ring at rest if it is kept such that its diameter is along the axis of an ideal solenoid carrying a current of 10 A and has $1000 \text{ turns/ length}$.
 (A) $4 \times 10^{-3} \text{ N-m}$ (B) 2 N-m
 (C) $\frac{\pi}{10} \text{ mN-m}$ (D) $\frac{\pi}{5} \text{ mN-m}$

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a given series LCR circuit as shown in figure. There are 10 switches. Answer the following questions.



$$E_1 = E_0 \sin \omega t \quad E_2 = E_0 \sin(\omega t + \phi)$$

9. Switch S_8, S_9 & S_{10} are closed. Now it's observed if rest switches are opened or switches S_3, S_4 are closed (rest opened) effective value of current in the circuit is same. The value of X_C is :
- (A) R (B) $2R$
 (C) $\frac{R}{2}$ (D) $\frac{R}{3}$
10. Switch S_8, S_6, S_7 are closed. Now effective current if S_1, S_2 closed (Rest opened) is I_1 and if S_3, S_4 closed (Rest Opened) is I_2 value of $\left(\frac{I_1}{I_2}\right)^2$ is:
- (A) $\frac{3}{5}$ (B) $\frac{2}{3}$
 (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

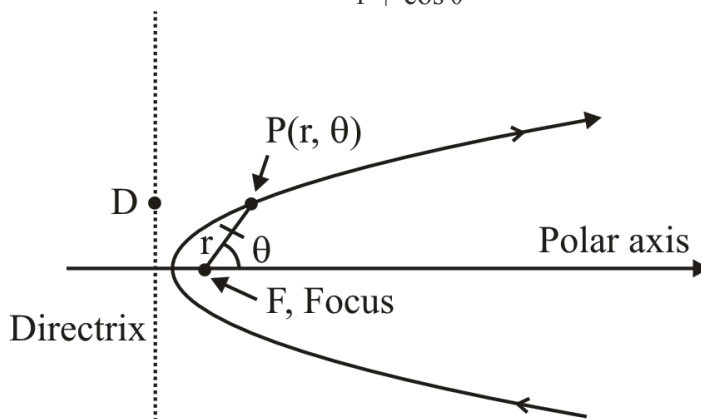
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider a conducting wire in the shape of a parabola whose equation is $y^2 = 4x$. It carries a current of 40 A flowing from the part in fourth quadrant to the part in first quadrant. The equation of parabola can also be written in terms of polar coordinates as $r = \frac{\ell}{1 + \cos \theta}$, where ℓ is the semi latus rectum.

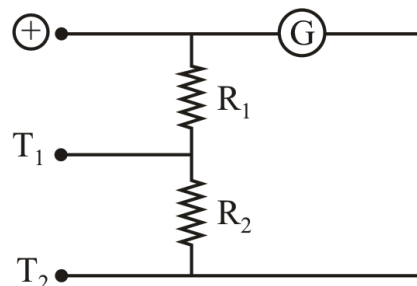
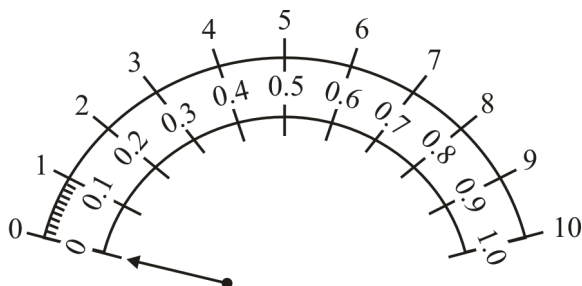


1. What is the magnetic field (in μT) at the focus due to the part of parabola between the latus rectum at the focus?
2. If a long current carrying wire is kept along the directrix and carries a current of 10 A, what is the force (in mN) due to the wire on the parabola?

Space for Rough Work

Paragraph for Questions 3 and 4

To enhance the sensitivity, an Ammeter is to be designed with two kinds of graduation on its scale 0 to 10 A and 0 to 1 A. For that a galvanometer of resistance $50\ \Omega$ and full scale deflection current $1\ \text{mA}$ was used along with two resistances R_1 and R_2 as shown. Either of T_1 or T_2 is to be used as negative terminal of the Ammeter.

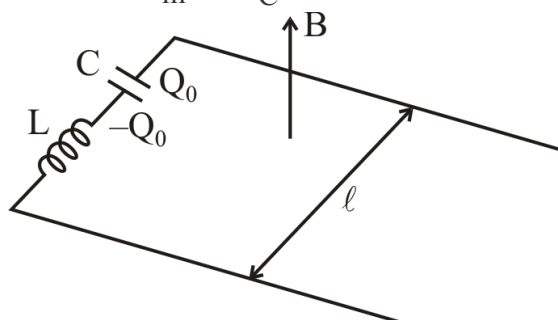


3. What is the value of R_1 (in milliohms)?
4. What is the value of R_2 (in milliohms)?

Paragraph for Questions 5 and 6

An inductor and a charged capacitor are connected to two long rails. A rod of mass m and length ℓ is gently kept on rails to complete the circuit. A uniform magnetic field is present in the entire space.

Resistance of circuit is negligible. Take $\frac{B^2 \ell^2}{m} = \frac{1}{C}$, $LC = 8\ \text{SI units}$, $Q_0 = 200\ \mu\text{C}$



5. What is the time period (in sec) of oscillation of rod thereafter?
6. What is the maximum current (in mA) in the circuit?

Space for Rough Work

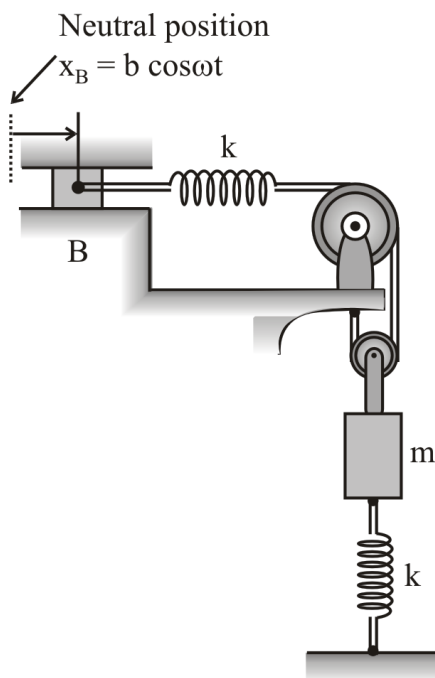
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. The spring attachment point B is given a horizontal motion $x_B = b \cos \omega t$. Determine the critical driving frequency ω for which the oscillations of the mass m tend to become excessively large. Neglect the friction and mass associated with the pulleys. The two springs have the same stiffness k . If $\omega = \sqrt{\frac{nk}{m}}$, fill value of n .



Space for Rough Work

8. A capacitor is uncharged at $t = 0$. It is charged through a variable current I .

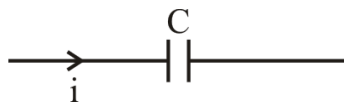
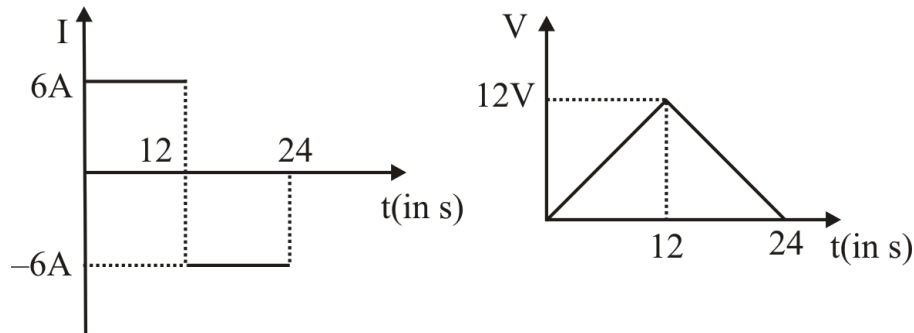
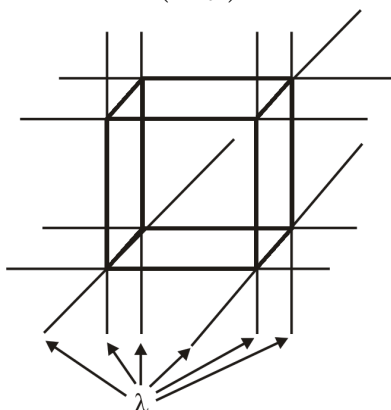


Figure shows variation of I with time and corresponding change in potential difference across capacitor with time. Find the capacitance (in farad) of capacitor used in the circuit.



9. Twelve infinite long wire of uniform linear charge density (λ) are passing along the twelve edges of a cube. If electric flux through any face of cube is $\left(\frac{\lambda \ell}{n\epsilon_0}\right)$. Then value of n is :



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

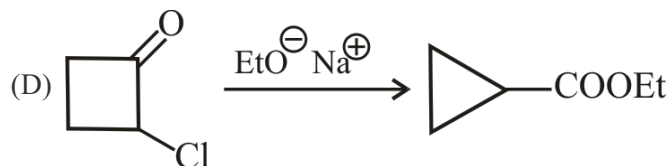
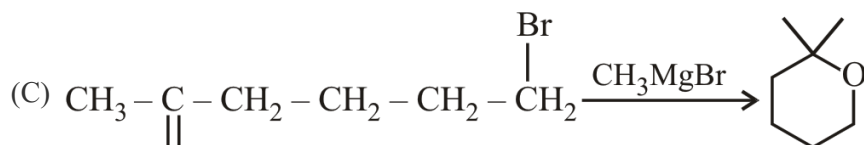
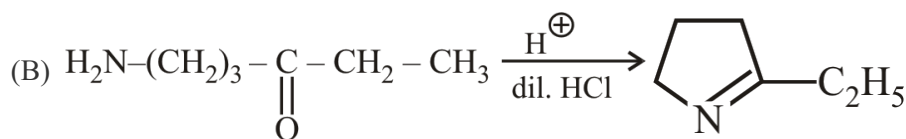
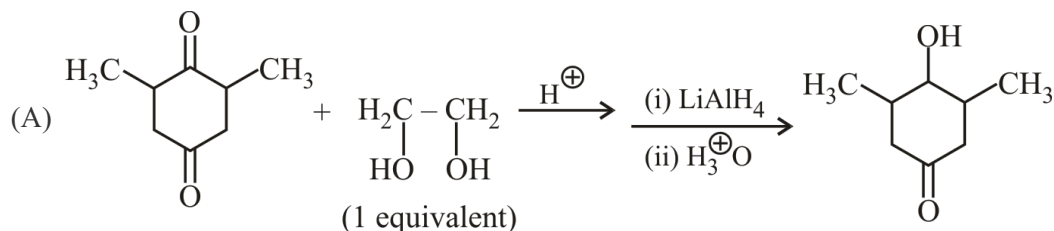
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

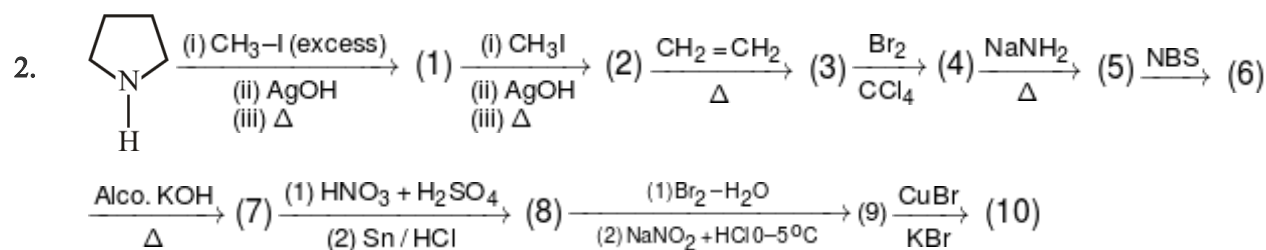
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

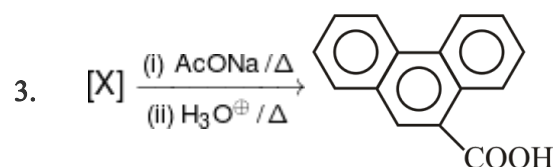
1. The correct product(s) is/are mentioned is





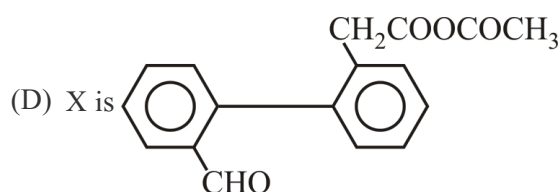
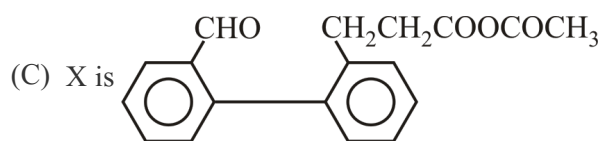
Choose the correct ones

- (A) Reaction involved in the formation of product (1) follows Hofmann elimination.
 (B) The number of Br atoms present in product (10) are four
 (C) Ethene is a dienophile in the reaction from (2) to (3)
 (D) The product (7) can also be obtained when aniline is treated with HNO_2 in cold followed by heating with H_3PO_2 .

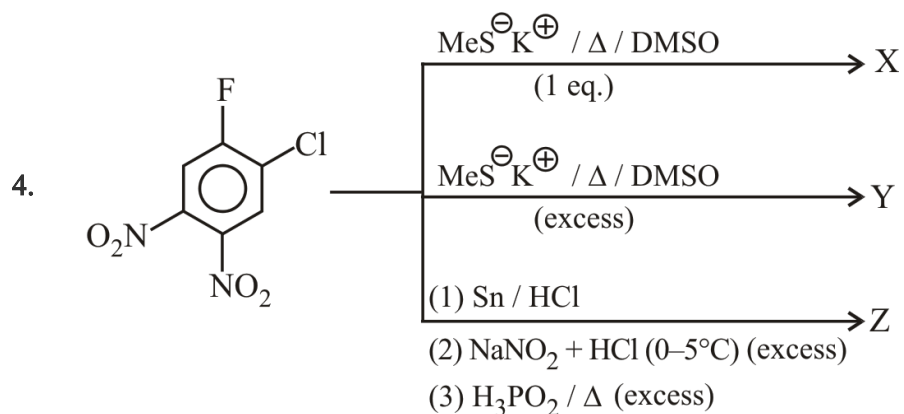


The correct statements regarding above reaction are

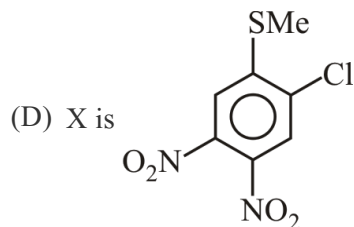
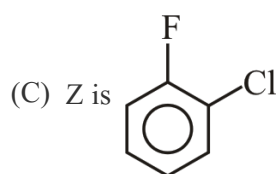
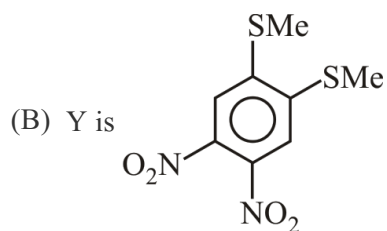
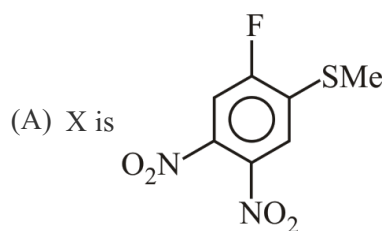
- (A) Compound X gives positive Tollen's test
 (B) Reaction proceeds through carbanion formation



Space for Rough Work

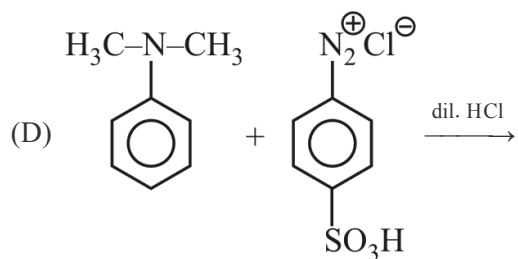
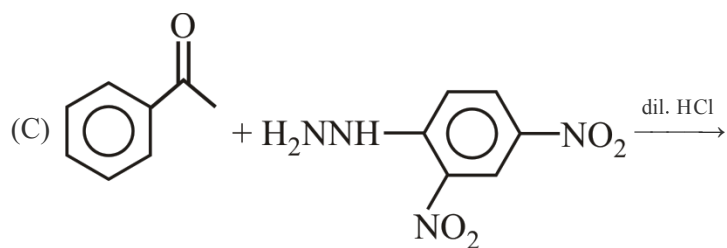
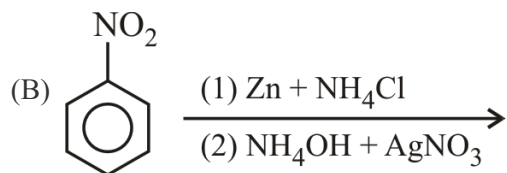
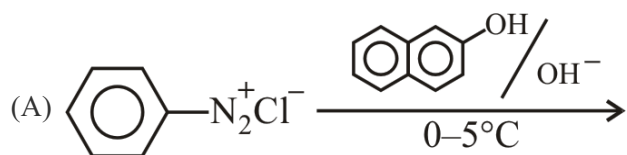


The correct option are

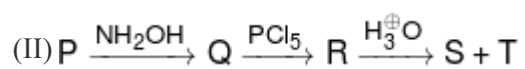
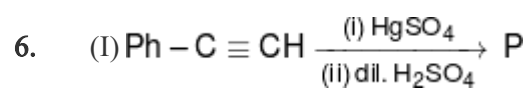


Space for Rough Work

5. How many of the following reactions will give yellow-orange-red (YOR) colour



Space for Rough Work



In the above sequence of reactions S + T may be

- (A) $\text{PhCOOH} + \text{CH}_3\text{-NH}_2$
- (B) $\text{PhCOOH} + \text{CH}_3\text{COOH}$
- (C) $\text{Ph-NH}_2 + \text{CH}_3\text{COOH}$
- (D) $\text{PhCH}_2\text{-CH} = \text{NOH} + \text{CH}_3\text{COOH}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

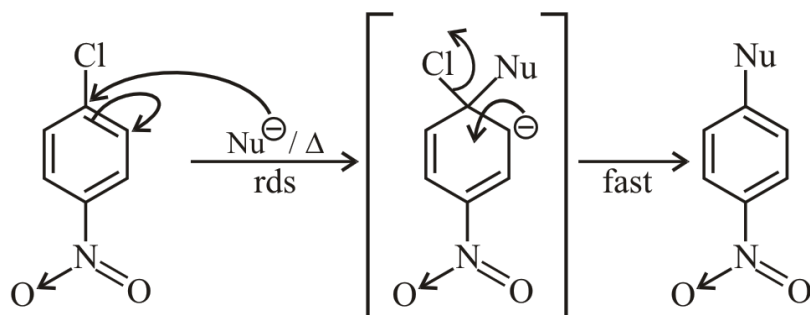
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

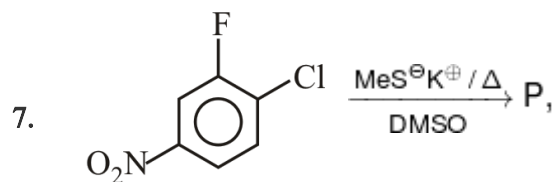
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

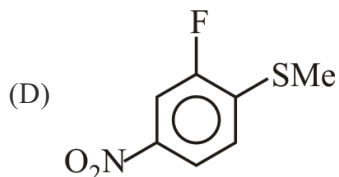
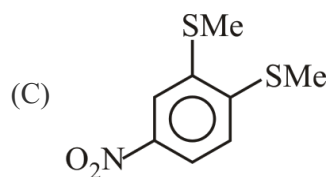
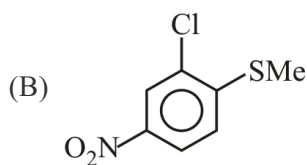
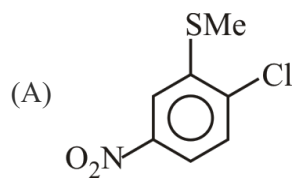
For nucleophilic substitution in halo benzene at least one strong electronwithdrawing group must be present at ortho or para position to leaving group. Reaction passes through a non-aromatic, resonance stable carbanion intermediate. Rate of reaction $\text{ArF} > \text{ArCl} > \text{ArBr} > \text{ArI}$.



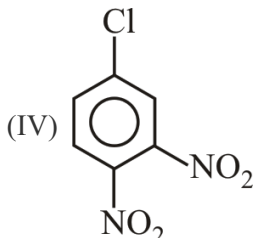
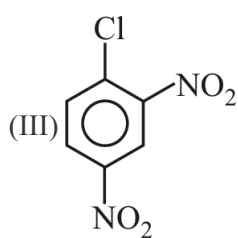
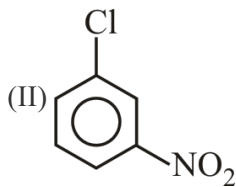
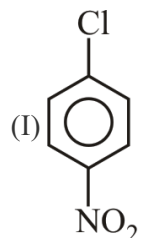
Space for Rough Work



The major product P is?



8. The order of rate of nucleophilic substitution reaction in the following is



(A) I > II > III > IV

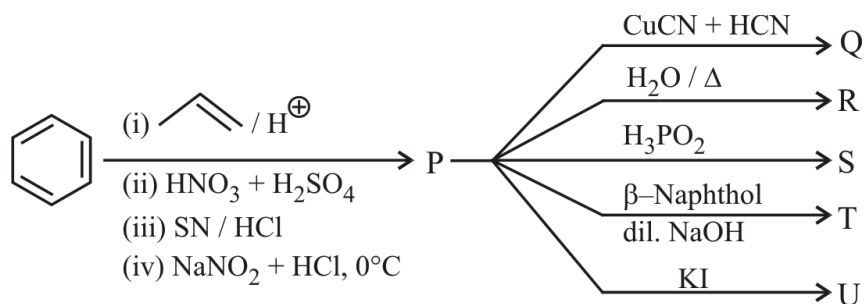
(B) III > IV > I > II

(C) IV > III > II > I

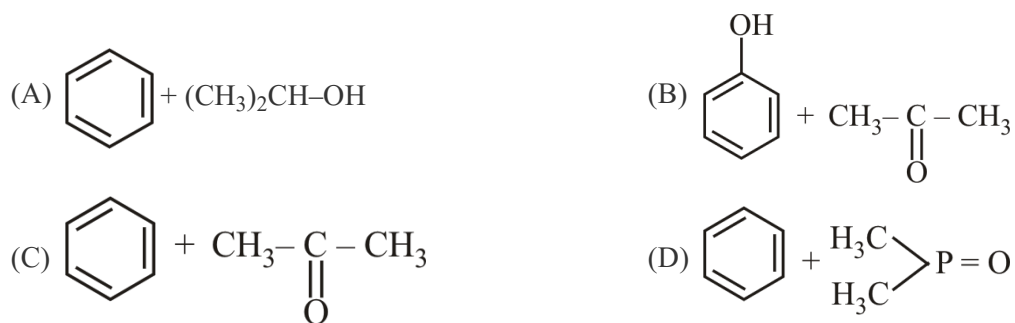
(D) III > I > IV > II

Space for Rough Work

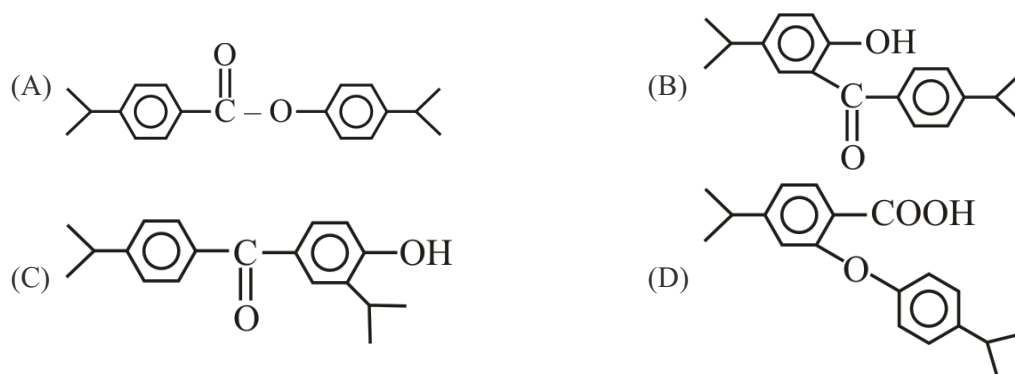
Paragraph for Questions 9 and 10



9. Compound S on reaction with O_2 / Δ followed by acid hydrolysis will give



10. $Q \xrightarrow[\text{(iii) } AlCl_3 / \Delta]{\text{(i) } H_3O^+}$
 (ii) R



Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

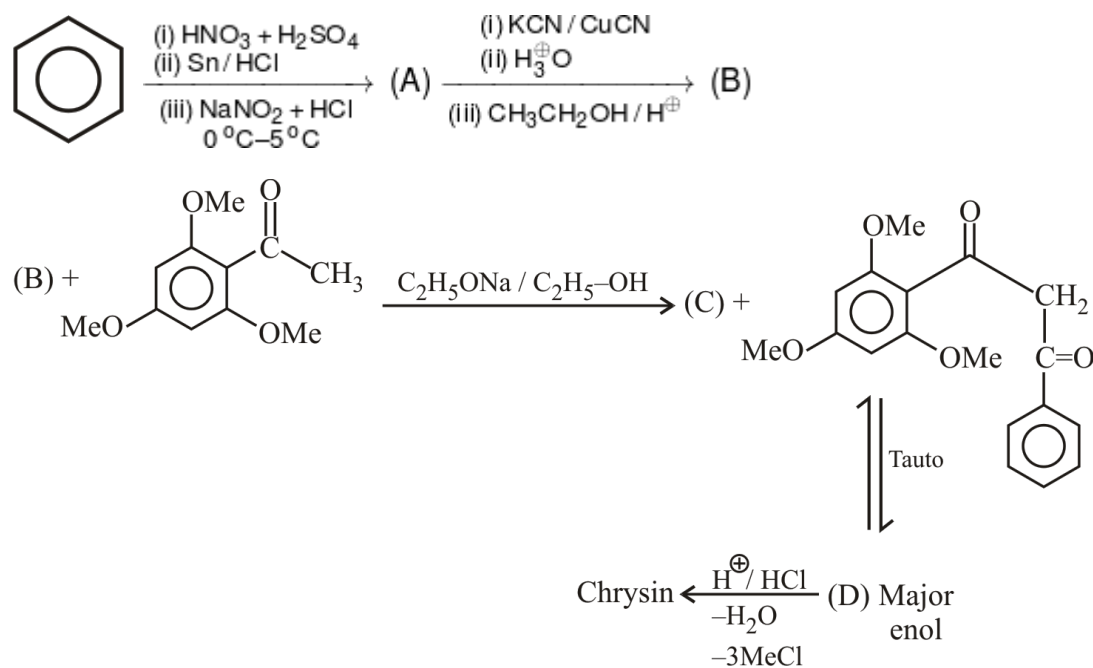
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

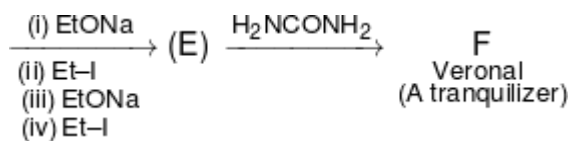
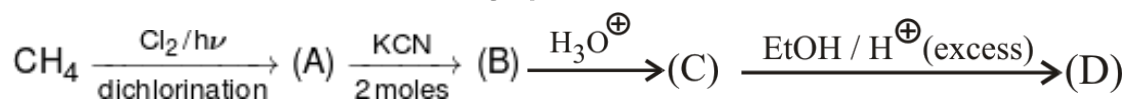
Chrysin is an organic compound, found in honey. It is used for bodybuilding for treating anxiety, inflammation, boldness etc. One of the methods of preparation of chrysin follows the reaction scheme :



1. The degree of unsaturation of chrysin is _____
2. The compound (A) in the above scheme on reaction with steam followed by treatment with $\text{CO}_2 / \text{NaOH}$, then with acetic anhydride gives a medically important compound which contains atoms of oxygen.

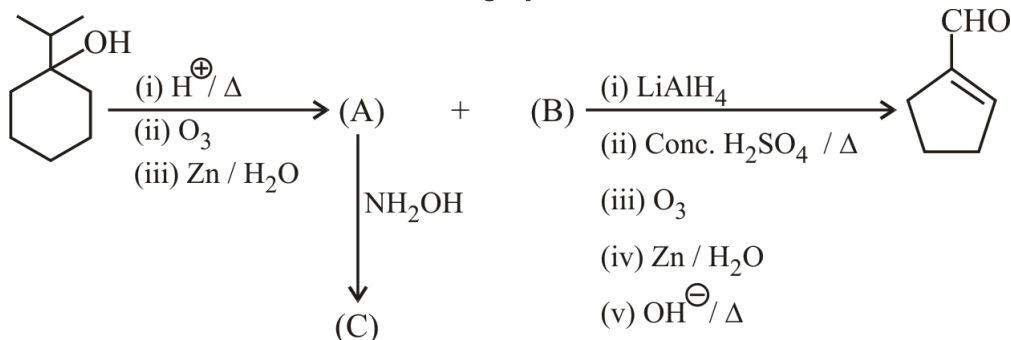
Space for Rough Work

Paragraph for Questions 3 and 4



3. How many sp^2 carbons are present in (F)
4. Compound (C) on heating gives a product in which percentage of oxygen is _____

Paragraph for Questions 5 and 6



5. The percentage of N in compound (C) is _____
6. Compound (B) on heating with dil. NaOH gives a product whose molecular weight is _____

Space for Rough Work

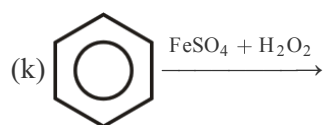
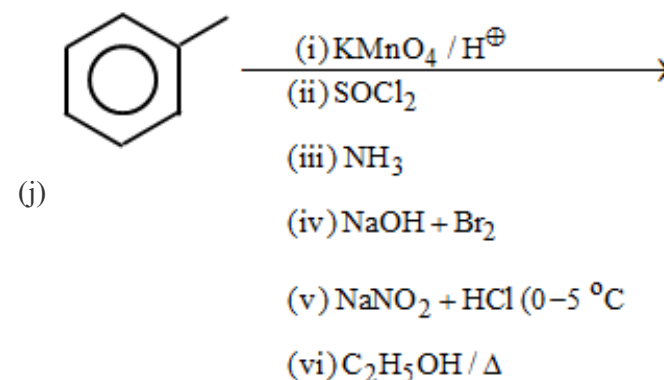
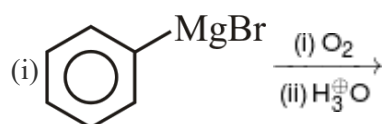
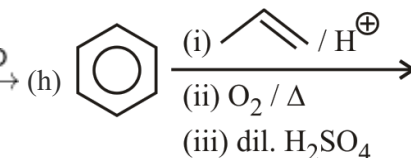
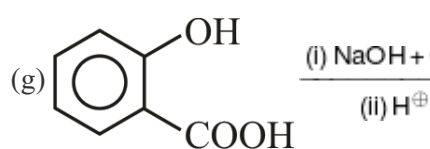
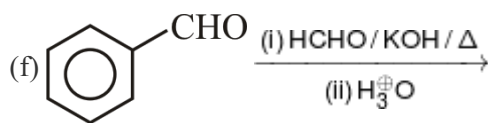
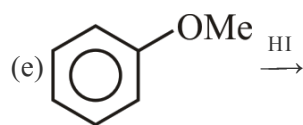
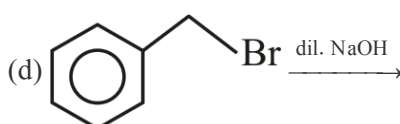
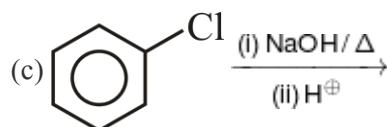
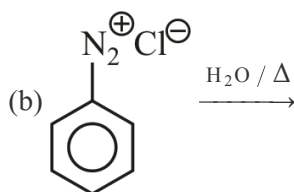
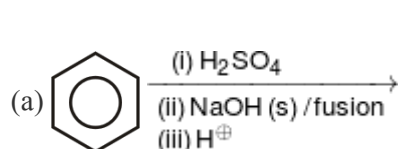
SECTION-II (ii) : (Maximum Marks: 12)

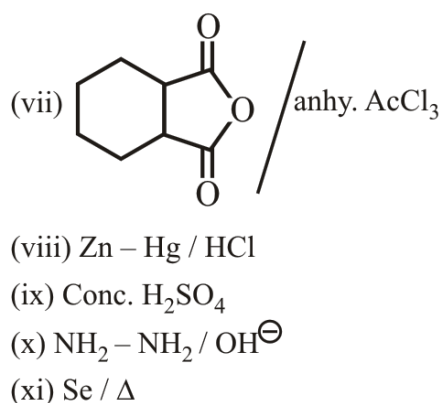
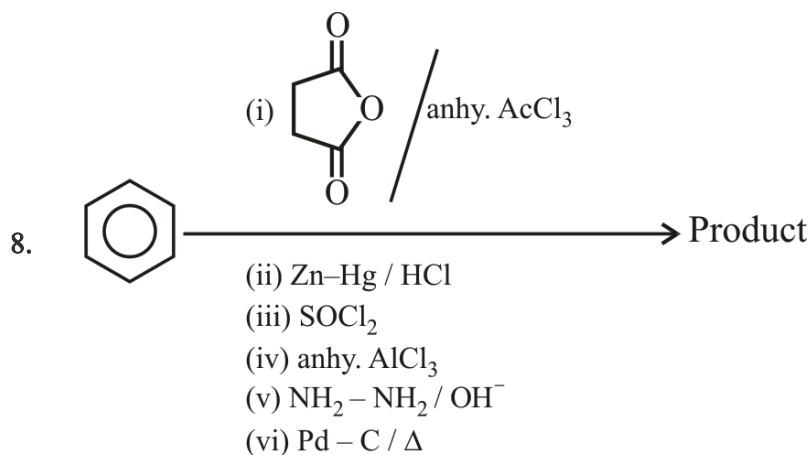
- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

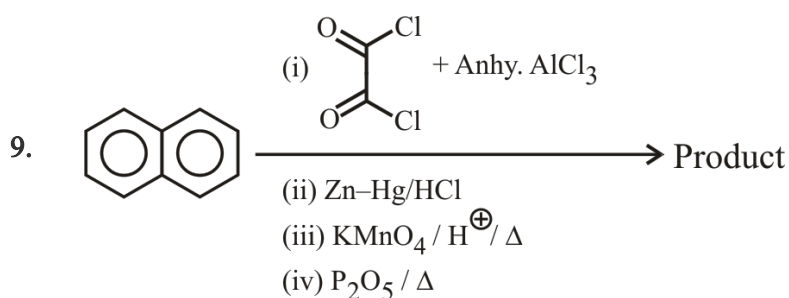
Zero Marks : 0 In all other cases

7. Identify the reaction which give phenol as the product





How many benzene rings will be present in the product?



The number of oxygen atoms present in the product is :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A parabola with focus (1, 0) and tangent at (4, 2) is $y = x - 2$ then which of the following is/are true:

(A) Equation of axis of symmetry is $3x - 2y = 3$

(B) Length of latus rectum is $\frac{2}{\sqrt{13}}$

(C) Equation of directrix is $2x + 3y = 1$

(D) Equation of parabola is $9x^2 + 4y^2 - 12xy - 22x + 6y + 12 = 0$

Space for Rough Work

2. Let PA and PB be the pair of tangents drawn to the hyperbola $25x^2 - 16y^2 = 400$ from the point P(4, -1), then
- (A) Equation of tangent from P is $x + 4 = 0$
- (B) Equation of tangent from P is $x - 4 = 0$
- (C) Equation of tangent from P is $13x + 4y = 48$
- (D) Area of ΔPAB is $\frac{1}{6}$ square unit
3. Let $(x - k)^2 + y^2 = r^2$, ($k, r > 0$) is the largest circle, no part of which is lying outside the ellipse $3x^2 + 4y^2 = 12$. Now
- (A) if $k < \frac{1}{2}$ then $r + k = 2$
- (B) if $k > \frac{1}{2}$ then $r + k = 2$
- (C) if $k < \frac{1}{2}$ then $r + k < 2$
- (D) if $k > \frac{1}{2}$ then $r + k < 2$
4. Given equation of ellipse E and circle C
- $$E \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 ; C \equiv (x - c)^2 + y^2 = c^2.$$
- If the ellipse E is of smallest possible area(Δ) such that it contains the circle C, then which of the following options is/are correct?
- (A) If $\sqrt{2}c = 1$, then $\sqrt{3}e = \sqrt{2}$
- (B) If $c = \sqrt{3}$, then $2\Delta = 9\sqrt{3}\pi$
- (C) If $\sqrt{2}c = \sqrt{3}$, then $\sqrt{3}e = \sqrt{2}$
- (D) If $\sqrt{2}c = \sqrt{3}$, then $2\Delta = 9\sqrt{3}\pi$

Space for Rough Work

5. Two hyperbolas H_1 and H_2 with eccentricities e_1 and e_2 are such that their latera recta form a rectangle. Then, which of the following options is/are correct?
- (A) $e_1 e_2 \geq 2$
- (B) If $e_1 = 2$, then $3e_2 = 1 + \sqrt{11}$
- (C) $e_1 e_2 \geq 3$
- (D) If $e_1 = 3$, then $16e_2 = 3 + \sqrt{265}$
6. Let P be a point with positive ordinate on ellipse $7x^2 + 16y^2 = 112$ with foci as S_1 and S_2 . If area of $\Delta S_1 P S_2$ is $3\sqrt{7}$ square units then radius of escribed circle of $\Delta S_1 P S_2$ may be :
- (A) $2\sqrt{7}$ (B) $\sqrt{7}$
- (C) $3\sqrt{7}$ (D) 4

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

For the origin O, given two points A and B such that $\angle AOB = 90^\circ$ on a parabola $y = x^2$. The line joining A and B has slope 2.

7. If pair of tangents are drawn from every point on the line AB to the parabola $y^2 = 4x$, then the chord of contact passes through a fixed point its co-ordinate is-
- (A) $\left(\frac{1}{2}, 3\right)$ (B) $\left(\frac{1}{2}, 1\right)$
- (C) $\left(\frac{1}{2}, 5\right)$ (D) (1, 5)
8. The area of ΔOAB is-
- (A) $\sqrt{2}$ (B) $\sqrt{3}$
- (C) $\sqrt{5}$ (D) $\sqrt{7}$

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a hyperbola with centre at origin. The line $x - \sqrt{3}y = 2$ is tangent to the hyperbola at $M(5, \sqrt{3})$ and intersects the asymptotes at A and B such that $AB = 6\sqrt{3}$ units.

9. Equation of asymptotes are -

(A) $15x^2 + 19y^2 + 24\sqrt{3}xy = 0$

(B) $15x^2 + 19y^2 + 14\sqrt{3}xy = 0$

(C) $19y^2 + 14\sqrt{3}xy - 15x^2 = 0$

(D) $15y^2 - 14\sqrt{3}xy + 19x^2 = 0$

10. Angle subtended by AB at the centre of hyperbola is-

(A) $\tan^{-1} \frac{6\sqrt{3}}{\sqrt{109}}$

(B) $\sin^{-1} \frac{4}{5}$

(C) $\sin^{-1} (6\sqrt{3})$

(D) $\cos^{-1} \frac{1}{\sqrt{109}}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider an ellipse $\frac{x^2}{4\alpha} + \frac{y^2}{\alpha} = 1$ (where α is a positive parameter) and parabola $y^2 = 8x$. If a tangent to the parabola meet the co-ordinate axes at A and B

1. If $\alpha = 8$, then the equation of line which touches both parabola and ellipse is $x + by + a = 0$, ($b < 0$), then value of $(a + b)$ is -
2. If two of the three normals drawn from $(a, 0)$ on the ellipse to the parabola $y^2 = 8x$ are perpendicular, then the value of ' α ' is

Space for Rough Work

Paragraph for Questions 3 and 4

Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a hyperbola such that its eccentricity and eccentricity of its conjugate hyperbola

are same. Let foot of the perpendicular drawn from focus of H to its one of the tangent is $\left(\frac{1}{2}, \frac{\sqrt{15}}{2}\right)$.

Tangents drawn at the ends of chord of length 2 unit of the circle $C : x^2 + y^2 = a^3 - 4b^2 + 12$, meets each other at 'P' on the transverse axis of H .

3. Let $E : \frac{x^2}{16} + \frac{y^2}{\beta^2} = 1$ ($\beta < 4$) be the ellipse which cuts H orthogonally then β^2 is-
4. If length of double ordinate of H passing through 'P' is L , then value of $3L^2$ is-

Paragraph for Questions 5 and 6

A conic C passing through $P(1,2)$ is such that the slope of its tangent at any point on the conic is inversely proportional to the ordinate of that point and conic C passes through origin.

5. If a circle touches the conic C at the point $P(1,2)$ and passes through the focus of the conic then its radius is 'K' then value of K^2 is -
6. If the slope of the normal from a point $(3,-1)$ to the conic C are m_1, m_2 & m_3 (real or imaginary) then the value of $\sum_{i=1}^3 \frac{1 - m_i}{1 + m_i}$ is-

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. From any point on the line $(t + 2)(x + y) = 1$, $t \neq -2$, tangents are drawn to the ellipse $4x^2 + 16y^2 = 1$. It is given that chord of contact passes through a fixed point. Then the number of integral values of 't' for which the fixed point always lies inside the ellipse is
8. The family of lines $\alpha x + y\sqrt{\alpha^2 - 1} = 1 \quad \forall$ permissible values of α touches a fixed hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If 'e' denotes the eccentricity of the hyperbola then $[e^5]$ (where $[.]$ denotes the greatest integer function) is
9. In hilly areas, reception of radio and television signals is sometimes poor. Consider an idealized case where a hill is represented by graph $y = x - x^2$, a transmitter is located at $(-1, 1)$ and a receiver is located on the other side of hill at $(x_0, 0)$. In order to get unobstructed reception of signal the closest the receiver can be to hill is $\frac{a\sqrt{b} - c}{d}$ (where a, b, c, d are prime numbers), then $(a + b + c + d)$ is
-

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work